

Metro Ticket Generating System

PROJECT DESIGN PHASE

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Team ID	T V SAI LAXMI
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

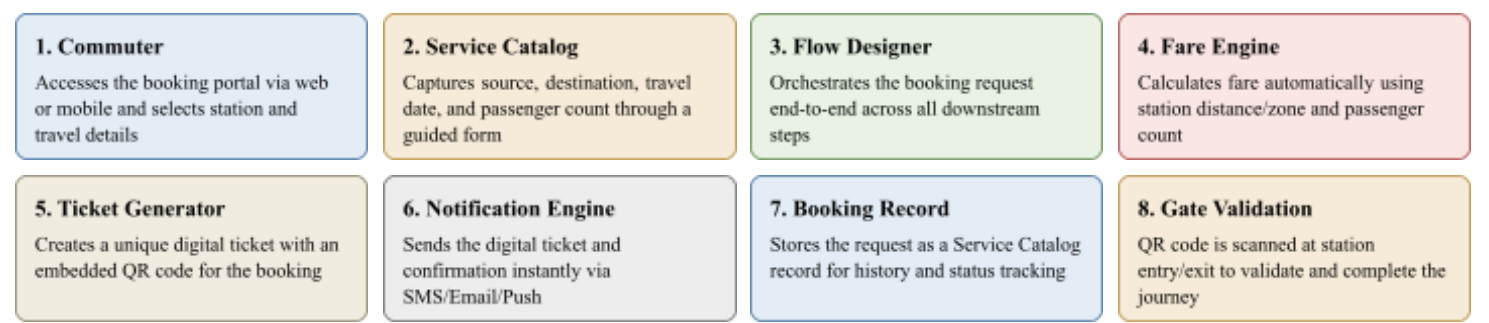
3. Solution Architecture

3.1 Architecture Overview

The Smart Metro Ticket Booking System is designed as a self-service workflow application built entirely on the ServiceNow platform. The commuter interacts with a Service Catalog booking form, which triggers a Flow Designer orchestration to calculate the fare, generate a digital QR ticket, notify the commuter, and log the booking record — all without any manual intervention from counter staff.

3.2 Architecture Diagram

End-to-End Booking Flow



Arrows in the numbered flow indicate the sequential order of processing, from booking request to gate validation.

3.3 Layered Architecture View

Layer	Component	Responsibility
Presentation Layer	ServiceNow Service Portal / Service Catalog widget	Provides the commuter-facing booking form and displays fare, ticket, and booking history
Process Layer	Flow Designer	Orchestrates fare calculation, ticket generation, notification, and record creation as a single automated flow
Business Logic Layer	Fare calculation script / decision table	Applies zone/distance-based fare rules and passenger count multipliers
Integration Layer	Notification module, QR generation utility	Generates the QR-coded ticket and pushes SMS/Email/Push notifications to the commuter
Data Layer	Service Catalog request tables (custom tables)	Persists booking details, fare, ticket status, and history for each commuter

3.4 Technology Components (ServiceNow)

ServiceNow Component	Used For
Service Catalog	Commuter-facing booking form with station, date, and passenger selection
Flow Designer	End-to-end workflow automation — fare calculation, ticket creation, notification, record logging
Notifications (Email/SMS/Push)	Instant delivery of the digital ticket and booking confirmation
Custom Tables	Storage of station master data, fare rules, and booking/ticket records
Decision Tables	Zone/distance-based fare rule evaluation
Service Catalog Request Records	Booking history and status tracking per commuter

3.5 Data Flow Description

- The commuter submits a booking request through the Service Catalog form with station and passenger details.

- Flow Designer receives the request and calls the fare calculation logic to compute the exact fare.
- On confirmation, a unique digital ticket record is created and a QR code is generated and linked to it.
- The Notification module sends the ticket and confirmation to the commuter's registered email/phone.
- The booking is stored as a Service Catalog request record, making it available under the commuter's booking history.
- At the station, the QR code is scanned at the gate to validate entry and exit for the journey.

3.6 Security and Scalability Considerations

- **Access control:** Role-based access ensures commuters can only view and manage their own bookings.
- **Ticket integrity:** Each QR code is uniquely tied to a single booking record to prevent duplication or misuse.
- **Scalability:** Built on Flow Designer and native tables, the solution can handle increasing booking volumes without redesign.
- **Extensibility:** The layered design allows future additions such as payment gateways, season passes, and chatbot support without disrupting the core booking flow.
- **Auditability:** Every booking, fare calculation, and notification is logged against the Service Catalog request record for traceability.